

DECT-2020 NR Link Distance Performance in Varying Environments: Models and Measurements

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Abstract—Digital Enhanced Cordless Telecommunications 2020 New Radio (DECT-2020 NR) has garnered recognition as an alternative for cellular 5G technology in the internet of things industry. This paper presents a study centered around the analysis of the link distance performance in varying environments for DECT-2020 NR. The study extensively examines and analyzes received signal strength indicator and resulting path loss values in comparison with theoretical models, as well as packet success rates (SR) and signal-to-noise ratio against varying distances. The measurements show that with an SR of over 90 %, an antenna height of 1.5 m, indoor link distances with a single device-to-device connection with 0 dBm transmission (TX) power can reach over 60 m in non-line-of-sight (NLOS) areas and up to 190 m in LOS areas with smaller -8 dBm TX power. Similarly, for outdoor use cases, link distances of over 600 m can be reached with +19 dBm TX power.

Index Terms—DECT-2020 NR, internet of things, radio propagation models, link-level measurements.

I. INTRODUCTION

In the current era of the internet, humanity has witnessed an unprecedented level of interconnections. The growth of limited-range networks and the expansion of devices connected to these networks have led to a continuous interconnection between devices. This interconnection enables devices to collect and distribute information through a series of communication processes, often involving minimal or no human intervention [1]. These devices are embedded with intelligence and communication capabilities, including sensors, cellphones, vehicles, household devices, and medical equipment. This transformation has led to the emergence of the internet of things (IoT), marking a significant transition from the current human-focused internet to a network where devices communicate with each other autonomously [2].

Digital Enhanced Cordless Telecommunications 2020 New Radio (DECT-2020 NR) is a new wireless communication standard designed specifically for industrial IoT applications by targeting massive machine-type communications (mMTC) and/or ultra-reliable low latency communications (URLLC) capabilities. It introduces a self-organizing, decentralized mesh topology, eliminating the necessity for dedicated infrastructure like base stations or a centralized core. Within this framework, individual devices can function as intermediary hubs, facilitating multi-hop data propagation when direct device-to-device links are unfeasible [3].

DECT-2020 NR standard has been incorporated into the 5G standards by International Telecommunication Union (ITU),

with European Telecommunications Standard Institute (ETSI) publishing its inaugural version in June 2020. Notably, the ITU Radiocommunication Sector (ITU-R) has encompassed DECT-2020 NR within the 5G NR standards under International Mobile Telecommunications (IMT) 2020, or IMT-2020, technology recommendation ITU-R M.2150-2 [4]. In line with this, the DECT-2020 NR standard, as outlined by ETSI, serves as a critical framework for achieving robust and seamless IoT device communication.

The radio interface architecture of DECT-2020 NR has four layers: physical layer, medium access control (MAC) layer, data link Control layer (DLC) with routing service and convergence layer (CVG). Physical layer comprises with two parts: synchronization training field (STF) and data field (DF). STF enables per-packet synchronization and the gain control at the receiver. DF section of the packet encompasses two distinct channels, namely physical control channel (PCC) and physical data channel (PDC) [5]. The PCC provides a control channel for MAC that can indicate short network ID, transmitter and receiver IDs of the transmission and used transmission parameters of the PDC. Transmission parameters include transmission power, modulation and coding scheme (MCS), transmission length, number of spatial streams, and redundancy versions. Furthermore, PCC encompasses hybrid automatic repeat request (HARQ) feedback channel quality indicator (CQI), and codebook index for MIMO transmissions. The PDC then contains the MAC PDU, containing the MAC protocol data and control units as defined in [5].

In this paper, radio propagation models are first presented and an extensive measurement campaign is performed to compare the achieved link distances with theoretical ones. The aim is to show how well DECT-2020 NR point-to-point connections, or in mesh network terms one-hop connections, are able to perform in numerous different environments with good packet success rates. These environments include multiple indoor and outdoor locations with both line-of-sight (LOS) and non-line-of-sight (NLOS) connections.

II. RADIO PROPAGATION MODELS

A. Propagation Models

Radio wave propagation models, which typically depict the radio environment as a function of frequency and distance between the terminals, are utilized to predict the path losses of DECT-2020 NR link level connections. The models considered for comparison with the measurements are the following ones:

1) *Free Space Path Loss Model*: The basic and most well-known model for the attenuation of the transmitted signal's power between two antennas through free space with a direct line of sight (LOS) between the transmitter and receiver is called the free space path loss (FSPL). The FSPL equation can be written as [6]:

$$FSPL \text{ (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10} \left(\frac{4\pi}{c} \right), \quad (1)$$

where d is the distance between antennas (m), f is frequency of the signal (Hz), and c is speed of light (m/s).

2) *3GPP TR 38.901 Path Loss Model*: 3GPP technical report version 38.901 focuses on frequencies ranging from 0.5 GHz to 100 GHz. In this model, indoor office environments and shopping malls are referred to as InH, indoor factories containing heavy machinery, assembly lines, and varying densities of clutter are referred to as InF. The two most relevant equations for this study from this model are the following [7]:

$$PL_{\text{InH-LOS}} \text{ (dB)} = 32.4 + 17.3 \log(d) + 20 \log(f_c), \text{ and} \quad (2)$$

$$PL_{\text{InF-LOS}} \text{ (dB)} = 31.84 + 21.50 \log(d) + 19 \log(f_c), \quad (3)$$

where d is defined as the distance between the antenna and the measurement point (m) and f_c is the carrier frequency (GHz).

3) *Two-Ray Ground Reflection Model*: Two-ray ground-reflection model accounts for both the direct signal path and the signal reflected off the ground. The equation stands as below [8]:

$$PL \text{ (dB)} = 40 \log_{10}(d) - 10 \log_{10}(G h_t^2 h_r^2), \quad (4)$$

where d is the distance between the transmitter (TX) and receiver (RX), h_t is the height of the TX, h_r is the height of the RX, and G is the combined TX-RX antenna gain.

4) *Okumura-Hata Model*: The Okumura-Hata model is particularly well-suited for urban environments with numerous buildings with relatively few tall obstructions. The model is formulated as follows[9]:

$$L_U \text{ (dB)} = 69.55 + 26.16 \log_{10}(f) - 13.82 \log_{10}(h_B) - C_H \\ + [44.9 - 6.55 \log_{10}(h_B)] \log_{10}(d). \quad (5)$$

where L_U is PL in urban areas (dB), f is the utilized frequency (MHz), h_B is the height of the base station antenna (m), C_H is antenna height correction factor, and d is the distance between the base station and mobile station (km).

5) *COST-231 Hata Model*: The COST-231 Hata model is applicable in the frequency range of 500 MHz to 2000 MHz. The basic equation for PL in dB is given by [10]:

$$PL \text{ (dB)} = 46.3 + 33.9 \log_{10}(f) - 13.82 \log_{10}(h_b) - \alpha h_m + \\ [44.9 - 6.55 \log_{10}(h_b)] \log_{10}(d) + C_m, \quad (6)$$

where f is the utilized frequency (MHz), d is the distance between access points (AP) and customer premises equipment (CPE) antennas (km), h_b is the AP antenna height above ground level (m), h_m is the CPE antenna height above ground

level (m), αh_m is a city size dependent variable, and C_m is 0 dB for suburban or open environments and 3 dB for urban environments.

B. Empirical Path Loss

Empirical PLs were calculated using the formula below to compare with the theoretical PL values for the corresponding scenarios. The empirical PL formula is defined as:

$$PL_{\text{emp}} \text{ (dB)} = P_{\text{tx}} - P_{\text{rx}} + G_{\text{tx}} - L_{\text{tx}} + G_{\text{rx}} - L_{\text{rx}}, \quad (7)$$

where P_{tx} is the transmission power of TX (dBm), P_{rx} is the receiving power of RX (dBm), G_{tx} is the TX antenna gain (dB), L_{tx} is the TX side connector losses (dB), G_{rx} is the RX antenna gain (dB), and L_{rx} is the RX side connector losses (dB).

III. DATA ACQUISITION AND ASSESSMENT

A. Equipment Details

For analyzing the propagation effects, pre-commercial devices have been utilized. These devices, model nRF9131 as shown in Fig. 1, were manufactured by Nordic Semiconductor and support the new DECT-2020 NR standard [11]. Nordic Semiconductor's proprietary software was employed to command and control these devices. Once the devices were set up for a link distance measurement, data was collected and stored for post-processing.

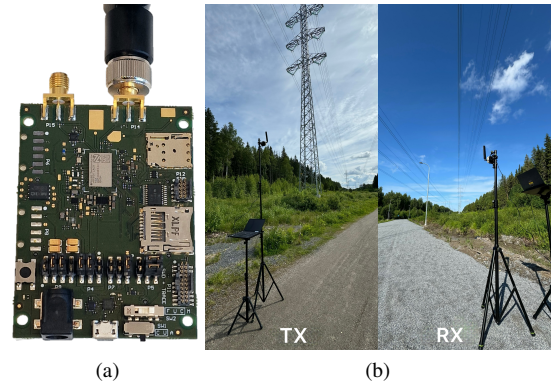


Fig. 1. a) The utilized nRF9131 board and b) Hallila power line measurement scenario.

B. Measurement Setup

A point-to-point connection was employed for measuring average received signal strength indicator (RSSI), path loss (PL), signal-to-noise ratio (SNR) and success rate (SR) by collecting 300 measurement samples (one sample per second) at each location point. Both the TX and RX were mounted on separate tripods to achieve stationary positions and desired heights. Moreover, antennas were connected to both of the devices as shown in Fig. 1. These Pulselarsen W5095X antennas are vertically polarized, omnidirectional blade antennas.

For all indoor distance measurements, a laser distance meter was employed to precisely measure the distance between the

TX and RX devices. In outdoor scenarios, online mapping tools were utilized to determine the distances between the devices by utilizing GPS coordinates measured at both ends with a common smart phone.

RSSI data from the RX, along with cyclic redundancy check (CRC) success and failure counts, were independently recorded and stored. Lastly, MATLAB software was employed to process the data from these files.

C. Data Assessment Procedure

1) *Received Signal Strength Indicator and Signal to Noise Ratio Calculation:* Following the acquisition of each dataset, the RSSI levels for both PCC and PDC were systematically organized. MATLAB script was subsequently employed to compute key statistical parameters, including the mean, standard deviation, maximum, and minimum values. The RSSI and SNR measurements, initially recorded in logarithmic scales, underwent a conversion process to linear scale to facilitate average value calculations, before being reconverted to logarithmic scale. This methodical approach was applied to the raw data (RSSI and SNR data sets) for each measured distance, ensuring a comprehensive representation of signal characteristics across varying spatial parameters. Furthermore, this average RSSI value in logarithmic form is subtracted from the transmitted power, while internal and connector losses along with TX-RX gains (total 1 dB per side) are added to compute the empirical PLs, according to (7). The measurements were conducted using omnidirectional antennas, which closely align with the assumptions made in the propagation models considered in this study.

2) *Success Rate Analysis:* In the measurement process, the device utilized a continuous reception method without implementing any re-transmission mechanism for packet error correction. Instead, CRC was employed as the primary means of error detection. The ‘CRC Okay Count’ served as central metric, enabling the quantification of successfully received packets that passed the error detection process. SRs were quantified as percentages, utilizing formulas below:

$$SR_{PCC} = \left(\frac{\text{PCC CRC Okay Count}}{\text{Request Count}} \right) \times 100 \quad (8)$$

$$SR_{PDC} = \left(\frac{\text{PDC CRC Okay Count}}{\text{Request Count}} \right) \times 100 \quad (9)$$

Finally, RSSI values, both theoretical and empirical PL equations and SRs were visualized through distinct curves plotted against distance. This meticulous data collection and analysis approach provided in-depth insights into the performance characteristics and reliability of the studied devices, while also offering valuable information about their applicability in various indoor and outdoor scenarios.

D. Data Collection from Measurement Scenarios

Link distance measurements were carried out at various sites situated at Tampere, Finland, to collect the data. These sites can be categorized as indoor and outdoor locations. Indoor

measurements were conducted at Tampere University Hervanta campus (buildings “Konetalo” and “Rakennustalo”) and Valmet Oy industry premises. Outdoor locations included various sites and areas around Tampere University, Ahvenisjärvi, Opiskelijankatu, Hallila, Kangasala, and Kaukajärvi, encompassing urban, sidewalk, forested, and lakeside environments. The general and DECT-2020 NR specific parameters considered for the measurements are shared in Table I.

TABLE I
CONSIDERED PARAMETERS FOR DECT-2020 NR LINK-LEVEL MEASUREMENTS

Parameter	Value
Payload length (in slots)	4
Modulation and Coding Scheme (MCS)	1 & 4
Successive Packet Transmission Interval	1 s
DECT-2020 Channel	1677 & 1667
TX Power	-20 dBm, 0 dBm, +19 dBm
Node Height above Ground Level	1.5 m
Antenna Gain	1.07 dB $\pm$ 1 dB
Carrier Frequency	1899 MHz
Channel Bandwidth	1.728 MHz

IV. RESULTS AND ANALYSIS

To ensure reliable communication and satisfactory performance of the devices under examination, the SR is required to exceed 90 %. This is considered a good threshold for a well-performing single-link connection when considering only one-hop and targeting maximum distance, without introducing an extensive amount of re-transmission in the link. In fact, 10 % packet error rate is the typical assumption in wireless data networks such as Long Term Evolution (LTE) [12]. Although a typical DECT-2020 NR mesh network is expected to consist of a large number of nodes enabling longer end-to-end distances with multiple hops, it can still guarantee up to 100 % SR by enabling link-specific re-transmission using HARQ for any packet error occurring in a link.

In order to help to analyse and visualize the results, nonlinear least squares (NLS) curve fitting was also applied to the measurement data for logarithmic scale PL values versus distance graphs. This was utilized, when several distances were measured in one location.

A. Minimum Transmission Power Requirements

A crucial parameter for long-term device deployment is minimizing TX power while maximizing range, which optimizes battery life. Additionally, utilizing optimum TX power reduces interference in the spectrum, improving system capacity and co-existence with other systems [13]. To elucidate these power-distance relationships across various propagation scenarios, Table II and Table III present a comprehensive summary of the minimum power requirements necessary to achieve maximum communication distances under the condition of reliable communication.

TABLE II
MINIMUM TRANSMISSION POWER REQUIREMENTS FOR RELIABLE
COMMUNICATION AT MAXIMUM DISTANCE IN INDOOR SCENARIOS

Location	Propagation Type	Devices Located at / Separated By	Maximum Distance Measured (m)	Minimum TX Power Required (dBm)
Konetalo 2nd Floor (1)	LOS	Straight Corridor	40	-20
Rakennustalo 2nd Floor (2)	LOS	Straight Main Corridor	120	0
Valmet Office (3)	LOS	Straight Main Corridor	137	0
Valmet Factory (4)	LOS	Long Factory Area	190	-8
Konetalo Inter-floors	NLOS	Concrete Floors (three floors)	10.71	0
Valmet Office	NLOS	Brick Wall & Glass	61	0
Valmet Office	NLOS	Brick Wall, Glass and Cubicles	61	0
Valmet Office	NLOS	Circular Main Corridor	111	+19

TABLE III
MINIMUM TRANSMISSION POWER REQUIREMENTS FOR RELIABLE
COMMUNICATION AT MAXIMUM DISTANCE IN OUTDOOR SCENARIOS

Location	Propagation Type	Devices Located at / Separated By	Maximum Distance Measured (m)	Minimum TX Power Required (dBm)
Kampusareena and Main Building Outdoors (5)	LOS	Free-Space	40	-20
Ahvenisjärvi Football Ground (6)	LOS	Free-Space	205	-8
Ahvenisjärvi Forest (7)	NLOS	Forest and Lake	265	+16
Opiskelijankatu Sidewalk (8)	LOS	Walking Bridge	610	+19
Hallila Power Line (9)	LOS	Free-Space	650	+19
Kaukajärvi Lake	LOS	Lake	2294	+19
Kangasala Observation Tower	LOS	Terrain	2470	+19

B. PL versus Distance

Fig. 2 illustrates empirical PL against distance for various indoor corridor LOS scenarios. Konetalo 2nd floor corridor consistently has the lowest PL, staying below 3GPP TR 38.901 InH-LOS threshold. Valmet factory corridor shows highest PL among all, but exhibits a moderate PL compared to InF-LOS model.

Moreover, Fig. 3 illustrates PL and distance relationship for various environments at outdoors LOS for 19 dBm TX power. The heights of the TX and RX have been designated as 10 m and 1.5 m respectively for the theoretical models. Ahvenisjärvi football ground showed a very close performance with the FSPL curve, whereas Opiskelijankatu sidewalk exhibited one of the highest PLs indicating impact of multipath propagation due to pedestrians and live vehicular traffic as well as more vegetation along the road. Kampusareena-Mainbuilding outdoor scenario showed notable variation in PL values as the devices are located between two buildings and Kampusareena is metal coated. Therefore, the signal got reflected very well from the buildings and showed tunneling effect in guiding the signal well between the closely located buildings. Under Hallila power lines, the link distance is very similar to the Opiskelijankatu sidewalk case. Ahvenisjärvi forest PL remained highest in comparing with other scenarios of the same distance due to the presence of a thick forest. However, all measured scenarios consistently showed lower PL than the Okumura-Hata and COST-231 Hata models predict at all distances and higher PL than FSPL model predicts after 100 m distance. The outdoor LOS measurements were performed with 19 dBm TX power to assess device performance at maximum distances with different heights, as shown in Table IV. Kangasala observation tower outperformed other sites, but this is due to its considerable height advantage (TX-RX height difference is 52.6 m). Excluding height differences, Kaukajärvi lake showed favorable propagation conditions over water, as one might expect. All measured scenarios exhibited significantly lower PL than the two models Okumura-Hata and COST-231 Hata predicted, which is due to the more pessimistic nature of these models.

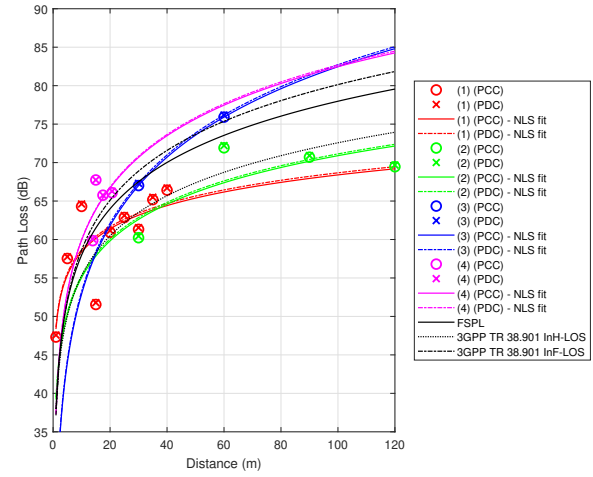


Fig. 2. PL versus distance in indoor scenarios for TX = -20 dBm. Site labels correspond to Table II locations. RSSI standard deviation: 0.34–4.17 dB.

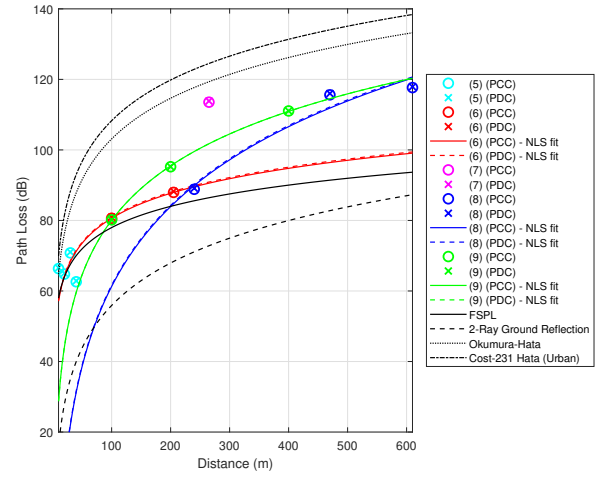


Fig. 3. PL versus distance in outdoor scenarios for TX = +19 dBm. Site labels correspond to Table III locations. RSSI standard deviation: 0.59–2.75 dB.

TABLE IV
PL FOR DIFFERENT (MAXIMUM) LINK DISTANCES AT OUTDOORS.

Scenario	Distance (m)	Height Difference (m)	Empirical PL PCC (dB)	Empirical PL PDC (dB)	FSPL (dB)	2-Ray Model PL (dB)	Okumura Hata Model PL (dB)	COST-231 Hata Model PL (dB)
Hallila Power Line	650	11.58	110.56	110.84	99.55	91.63	139.44	145.60
Kaukajärvi Lake	2294	1	120.38	120.61	105.16	115.00	-	-
Kangasala Tower	2470	52.6	113.70	113.96	105.83	96.93	144.50	149.67

C. SR & SNR versus RSSI

In this section, SR and SNR are simultaneously analyzed as a function of RSSI. From Fig. 4 and Fig. 5, it can be summarised that across all indoor and outdoor environments, SR consistently exceeded 90 % when RSSI values remained above -90 dBm for indoors and above -95 dBm for outdoors, demonstrating the capability to perform well even with weaker signal power levels.

The comparison of SNR to RSSI shows a more gradual increase compared to the sharp rise in the case of SR versus

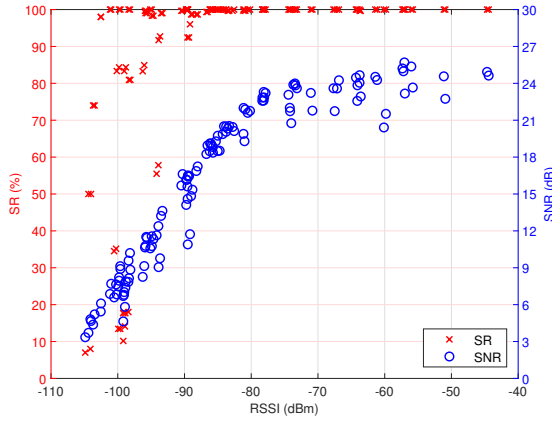


Fig. 4. SR and SNR as a function of RSSI at all indoor scenarios.

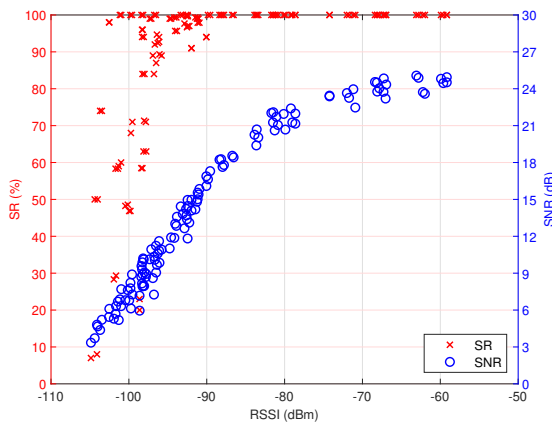


Fig. 5. SR and SNR as a function of RSSI at all outdoor scenarios.

RSSI. Generally, SRs surpassed 90 % when SNR values reached around 11 dB to 12 dB for indoors and around 12 dB to 15 dB for outdoors. SR data points cluster heavily near 100 % for stronger RSSI values, indicating excellent performance, while SNR values spread more, especially at stronger RSSI levels.

V. CONCLUSIONS

This paper presented a measurement-based study for practical DECT-2020 NR link distances in indoor and outdoor scenarios with pre-commercial equipment. The maximum one-hop link distance communication ranges were determined for diverse outdoor settings, including open fields, forests, urban areas, terrains and over water surfaces.

The measurements show that with an SR of over 90 % and TX power ranging from -20 dBm to 0 dBm, the indoor LOS distances range from 40 m to 190 m and for NLOS use cases a distance of over 60 m can be reached with 0 dBm TX power. For outdoor use cases, with the maximum TX power of 19 dBm, link distances of around 600 m can be reached in LOS environment with low device heights. With deploying devices higher e.g. light poles, roofs or masts, significantly longer

distances can be achieved as demonstrated by measurements done at Kangasala observation tower.

The indoor scenarios PL characteristics predominantly aligned with the 3GPP TR 38.901 InH-LOS model, which can be stated as the most suitable model for the measurements. The predominant deployment of devices at similar heights in outdoors indicates the need for a more sophisticated theoretical model to be considered as the most suitable model.

With the achieved link distance results, it can be concluded that indoor networks can be realized with small TX powers enabling, for example, battery operated networks to function for a long time. Furthermore, large outdoor areas can be covered with suitable link distances, that is, having up to several hundred meters between the nodes, when node installation height is only 1.5 m, enabling reduction of network installation and maintenance cost compared to high mass deployments typically used in LPWAN or cellular technologies.

The results of this study advance both device performance under DECT-2020 NR communication protocols while providing insights into IoT-enabled systems' propagation characteristics. The findings not only show the devices' readiness for industrial deployment in smart metering, electricity network QoS monitoring and industrial facility condition monitoring, as well as smart building and smart city applications, but also establish benchmarks for evaluating performance across various environmental conditions.

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